

Confidence Intervals & Hypothesis Testing



Constructing a confidence interval

- 1 A random sample of 50 students has a mean study time of $\bar{x} = 6.2$ hours per week, with sample standard deviation $s = 1.8$ hours. Construct a 95% confidence interval for the population mean study time, using a critical value of $z^* \approx 1.96$.
 - 2 Interpret the confidence interval from the previous problem in the context of the study time question, using correct statistical language.
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Margin of error and sample size

- 3 Explain two ways to decrease the margin of error of a confidence interval, and state a tradeoff involved with each.
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The logic of hypothesis testing

- 4 State the null hypothesis H_0 and alternative hypothesis H_a for a test of whether a coin is biased toward heads (true proportion of heads p), and classify the test as one-tailed or two-tailed.
 - 5 Explain what a p -value represents, in the context of a hypothesis test.
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Four-step hypothesis testing procedure

- 6 A manufacturer claims their light bulbs last on average $\mu = 1000$ hours. A sample of 40 bulbs has mean lifetime $\bar{x} = 985$ hours with sample standard deviation $s = 45$ hours. State the hypotheses and compute the test statistic for testing whether the true mean lifetime is less than claimed.
 - 7 Using the test statistic $z \approx -2.11$ from the previous problem and a p -value of approximately 0.017, state a conclusion at the $\alpha = 0.05$ significance level.
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Type I and Type II errors, and power

- 8 In the light bulb test above, describe what a Type I error would mean in context, and what a Type II error would mean in context.
- 9 Explain what the power of a test measures, and state one way to increase it.

Confidence intervals for a proportion

- 10 A poll of 400 randomly selected voters finds that 220 support a candidate. Construct a 95% confidence interval for the true population proportion, using $z^* \approx 1.96$.

Determining sample size for a target margin of error

- 11 A researcher wants a margin of error of at most 0.03 for a 95% confidence interval for a population proportion, using a conservative planning value of $\hat{p} = 0.5$ (which maximizes required sample size). Find the minimum sample size needed, using $n = \left(\frac{z^*}{m}\right)^2 p(1-p)$ with $z^* = 1.96$.

One-sample t-test for a mean

- 12 A nutritionist claims a new granola bar has a mean of 200 calories. A sample of 15 bars has mean $\bar{x} = 206$ calories with sample standard deviation $s = 9$ calories. Since the population standard deviation is unknown and n is small, a t -test is used. Compute the test statistic $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$.
- 13 Explain what it means for a hypothesis test result to be "statistically significant," and why statistical significance does not always imply practical importance.
- 14 A researcher wants to test whether a new fitness program changes the average resting heart rate of participants from the population baseline of $\mu = 72$ bpm. A sample of 36 participants after the program has mean resting heart rate $\bar{x} = 68.5$ bpm with sample standard deviation $s = 10.5$ bpm. State the hypotheses (choosing a two-sided alternative), compute the test statistic, and using a p -value of approximately 0.11, state a conclusion at $\alpha = 0.05$.
- 15 Explain the relationship between a confidence interval and a two-sided hypothesis test at the same confidence/significance level — specifically, describe how you could use a 95% confidence interval alone to decide whether to reject $H_0 : \mu = \mu_0$ at $\alpha = 0.05$, without separately computing a p -value.
- 16 A city council is deciding whether to fund a new public transit line, and wants to survey residents to estimate the true proportion who would use it regularly. A preliminary small survey suggests around 35% support, but the council wants a formal survey with a margin of error no larger than 2.5% at 95% confidence. Using $n = \left(\frac{z^*}{m}\right)^2 p(1-p)$ with the preliminary estimate $p = 0.35$ and $z^* = 1.96$, find the required sample size, rounding appropriately.

- 17 A restaurant chain wants to test whether a new menu item's average preparation time is longer than the target of 8 minutes, since customers have complained about slow service. A random sample of 30 orders for the new item has a mean preparation time of $\bar{x} = 8.7$ minutes with sample standard deviation $s = 1.8$ minutes. State the hypotheses for this one-sided test, compute the test statistic, and using a p -value of approximately 0.019, state a conclusion at $\alpha = 0.05$ in the context of the restaurant's concern.
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Multiple choice — easy 15

- 18 A sample of $n = 121$ has mean $\bar{x} = 393$. A claimed population mean is $\mu_0 = 409$, with sample standard deviation $s = 62$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.
- a. 2.84
 - b. -2.34
 - c. -3.34
 - d. -2.84
- 19 A sample of $n = 49$ has mean $\bar{x} = 264$. A claimed population mean is $\mu_0 = 265$, with sample standard deviation $s = 75$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.
- a. -0.09
 - b. 0.41
 - c. 0.09
 - d. -0.59
- 20 A sample of $n = 25$ has mean $\bar{x} = 513$. A claimed population mean is $\mu_0 = 495$, with sample standard deviation $s = 75$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.
- a. -1.20
 - b. 0.70
 - c. 1.70
 - d. 1.20

- 21 A sample of $n = 121$ has mean $\bar{x} = 371$. A claimed population mean is $\mu_0 = 370$, with sample standard deviation $s = 13$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.
- a. 1.35
 - b. 0.35
 - c. -0.85
 - d. 0.85
- 22 A sample of $n = 25$ has mean $\bar{x} = 469$. A claimed population mean is $\mu_0 = 471$, with sample standard deviation $s = 51$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.
- a. -0.20
 - b. 0.30
 - c. -0.70
 - d. 0.20
- 23 A sample of $n = 121$ has mean $\bar{x} = 453$. A claimed population mean is $\mu_0 = 437$, with sample standard deviation $s = 40$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.
- a. 3.90
 - b. 4.90
 - c. 4.40
 - d. -4.40
- 24 A sample of $n = 16$ has mean $\bar{x} = 157$. A claimed population mean is $\mu_0 = 183$, with sample standard deviation $s = 34$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.
- a. -2.56
 - b. -3.06
 - c. 3.06
 - d. -3.56

- 25 A sample of $n = 49$ has mean $\bar{x} = 408$. A claimed population mean is $\mu_0 = 406$, with sample standard deviation $s = 15$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.
- a. 0.43
 - b. -0.93
 - c. 0.93
 - d. 1.43
- 26 A sample of $n = 64$ has mean $\bar{x} = 53$. A claimed population mean is $\mu_0 = 77$, with sample standard deviation $s = 51$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.
- a. -3.76
 - b. -3.26
 - c. -4.26
 - d. 3.76
- 27 A sample of $n = 49$ has mean $\bar{x} = 187$. A claimed population mean is $\mu_0 = 162$, with sample standard deviation $s = 65$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.
- a. 2.69
 - b. 3.19
 - c. -2.69
 - d. 2.19
- 28 A sample of $n = 81$ has mean $\bar{x} = 195$. A claimed population mean is $\mu_0 = 205$, with sample standard deviation $s = 58$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.
- a. 1.55
 - b. -1.55
 - c. -2.05
 - d. -1.05

- 29 A sample of $n = 121$ has mean $\bar{x} = 348$. A claimed population mean is $\mu_0 = 346$, with sample standard deviation $s = 28$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.
- a. 1.29
 - b. 0.79
 - c. 0.29
 - d. -0.79
- 30 A sample of $n = 144$ has mean $\bar{x} = 125$. A claimed population mean is $\mu_0 = 131$, with sample standard deviation $s = 78$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.
- a. 0.92
 - b. -1.42
 - c. -0.92
 - d. -0.42
- 31 A sample of $n = 64$ has mean $\bar{x} = 241$. A claimed population mean is $\mu_0 = 244$, with sample standard deviation $s = 25$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.
- a. -0.46
 - b. -1.46
 - c. -0.96
 - d. 0.96
- 32 A sample of $n = 81$ has mean $\bar{x} = 448$. A claimed population mean is $\mu_0 = 459$, with sample standard deviation $s = 59$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.
- a. -2.18
 - b. -1.18
 - c. -1.68
 - d. 1.68

Multiple choice — medium 15

- 33** A random sample of $n = 121$ has mean $\bar{x} = 180$ and sample standard deviation $s = 16$. Construct a 99% confidence interval for the population mean, using $z^* \approx 2.576$.
- a. (176.25, 181.75)
 - b. (175.25, 184.75)
 - c. (178.25, 183.75)
 - d. (176.25, 183.75)
- 34** A random sample of $n = 64$ has mean $\bar{x} = 93$ and sample standard deviation $s = 17$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.
- a. (88.83, 97.17)
 - b. (87.83, 98.17)
 - c. (90.83, 97.17)
 - d. (88.83, 95.17)
- 35** A random sample of $n = 36$ has mean $\bar{x} = 156$ and sample standard deviation $s = 23$. Construct a 90% confidence interval for the population mean, using $z^* \approx 1.645$.
- a. (149.69, 160.31)
 - b. (151.69, 162.31)
 - c. (148.69, 163.31)
 - d. (149.69, 162.31)

- 36 A random sample of $n = 49$ has mean $\bar{x} = 143$ and sample standard deviation $s = 41$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.
- a. (130.52, 155.48)
 - b. (133.52, 154.48)
 - c. (131.52, 152.48)
 - d. (131.52, 154.48)
- 37 A random sample of $n = 144$ has mean $\bar{x} = 154$ and sample standard deviation $s = 20$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.
- a. (150.73, 157.27)
 - b. (149.73, 158.27)
 - c. (150.73, 155.27)
 - d. (152.73, 157.27)
- 38 A random sample of $n = 121$ has mean $\bar{x} = 40$ and sample standard deviation $s = 11$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.
- a. (37.04, 42.96)
 - b. (38.04, 41.96)
 - c. (38.04, 39.96)
 - d. (40.04, 41.96)
- 39 A random sample of $n = 64$ has mean $\bar{x} = 112$ and sample standard deviation $s = 35$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.
- a. (103.42, 120.58)
 - b. (102.42, 121.58)
 - c. (103.42, 118.58)
 - d. (105.42, 120.58)

- 40 A random sample of $n = 16$ has mean $\bar{x} = 108$ and sample standard deviation $s = 7$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.
- a. (103.57, 112.43)
 - b. (106.57, 111.43)
 - c. (104.57, 111.43)
 - d. (104.57, 109.43)
- 41 A random sample of $n = 25$ has mean $\bar{x} = 199$ and sample standard deviation $s = 22$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.
- a. (190.38, 207.62)
 - b. (190.38, 205.62)
 - c. (189.38, 208.62)
 - d. (192.38, 207.62)
- 42 A random sample of $n = 49$ has mean $\bar{x} = 53$ and sample standard deviation $s = 21$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.
- a. (49.12, 58.88)
 - b. (46.12, 59.88)
 - c. (47.12, 58.88)
 - d. (47.12, 56.88)
- 43 A random sample of $n = 49$ has mean $\bar{x} = 154$ and sample standard deviation $s = 59$. Construct a 90% confidence interval for the population mean, using $z^* \approx 1.645$.
- a. (140.13, 167.87)
 - b. (139.13, 168.87)
 - c. (140.13, 165.87)
 - d. (142.13, 167.87)

- 44 A random sample of $n = 81$ has mean $\bar{x} = 107$ and sample standard deviation $s = 24$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.
- a. (101.77, 110.23)
 - b. (100.77, 113.23)
 - c. (101.77, 112.23)
 - d. (103.77, 112.23)
- 45 A random sample of $n = 36$ has mean $\bar{x} = 23$ and sample standard deviation $s = 51$. Construct a 90% confidence interval for the population mean, using $z^* \approx 1.645$.
- a. (9.02, 34.98)
 - b. (9.02, 36.98)
 - c. (8.02, 37.98)
 - d. (11.02, 36.98)
- 46 A random sample of $n = 25$ has mean $\bar{x} = 125$ and sample standard deviation $s = 53$. Construct a 90% confidence interval for the population mean, using $z^* \approx 1.645$.
- a. (107.56, 142.44)
 - b. (107.56, 140.44)
 - c. (106.56, 143.44)
 - d. (109.56, 142.44)
- 47 A random sample of $n = 16$ has mean $\bar{x} = 22$ and sample standard deviation $s = 56$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.
- a. (-3.44, 49.44)
 - b. (-5.44, 49.44)
 - c. (-5.44, 47.44)
 - d. (-6.44, 50.44)

Multiple choice — hard 15

- 48** A poll of 400 randomly selected voters finds that 180 support a candidate. Construct a 99% confidence interval for the true population proportion, using $z^* \approx 2.576$.
- a. (0.386, 0.484)
 - b. (0.416, 0.514)
 - c. (0.366, 0.534)
 - d. (0.386, 0.514)
- 49** A poll of 150 randomly selected voters finds that 37 support a candidate. Construct a 99% confidence interval for the true population proportion, using $z^* \approx 2.576$.
- a. (0.156, 0.307)
 - b. (0.136, 0.357)
 - c. (0.186, 0.337)
 - d. (0.156, 0.337)
- 50** A poll of 150 randomly selected voters finds that 85 support a candidate. Construct a 90% confidence interval for the true population proportion, using $z^* \approx 1.645$.
- a. (0.480, 0.653)
 - b. (0.500, 0.633)
 - c. (0.530, 0.633)
 - d. (0.500, 0.603)

- 51 A poll of 150 randomly selected voters finds that 112 support a candidate. Construct a 90% confidence interval for the true population proportion, using $z^* \approx 1.645$.
- a. (0.668, 0.825)
 - b. (0.688, 0.805)
 - c. (0.688, 0.775)
 - d. (0.718, 0.805)
- 52 A poll of 300 randomly selected voters finds that 143 support a candidate. Construct a 95% confidence interval for the true population proportion, using $z^* \approx 1.96$.
- a. (0.450, 0.533)
 - b. (0.420, 0.533)
 - c. (0.400, 0.553)
 - d. (0.420, 0.503)
- 53 A researcher wants a margin of error of at most 0.04 for a 99% confidence interval for a population proportion, using a planning value of $p = 0.30$. Find the minimum sample size needed, using $n = (z^*/m)^2 p(1 - p)$ with $z^* = 2.576$.
- a. 821
 - b. 870
 - c. 871
 - d. 921

- 54 A researcher wants a margin of error of at most 0.02 for a 90% confidence interval for a population proportion, using a planning value of $p = 0.70$. Find the minimum sample size needed, using $n = (z^*/m)^2 p(1 - p)$ with $z^* = 1.645$.
- a. 1421
 - b. 1420
 - c. 1371
 - d. 1471
- 55 A researcher wants a margin of error of at most 0.03 for a 90% confidence interval for a population proportion, using a planning value of $p = 0.50$. Find the minimum sample size needed, using $n = (z^*/m)^2 p(1 - p)$ with $z^* = 1.645$.
- a. 802
 - b. 752
 - c. 751
 - d. 702
- 56 A researcher wants a margin of error of at most 0.03 for a 95% confidence interval for a population proportion, using a planning value of $p = 0.40$. Find the minimum sample size needed, using $n = (z^*/m)^2 p(1 - p)$ with $z^* = 1.96$.
- a. 975
 - b. 1024
 - c. 1075
 - d. 1025

- 57 A researcher wants a margin of error of at most 0.02 for a 90% confidence interval for a population proportion, using a planning value of $p = 0.60$. Find the minimum sample size needed, using $n = (z^*/m)^2 p(1 - p)$ with $z^* = 1.645$.
- a. 1574
 - b. 1624
 - c. 1674
 - d. 1623
- 58 A hypothesis test about a manufacturer's claimed mean product lifetime produces a p -value of 0.047. Using a significance level of $\alpha = 0.01$, state the correct conclusion.
- a. The test is inconclusive and must be repeated
 - b. Fail to reject H_0 ; not enough evidence that the true mean lifetime differs from the claim
 - c. Accept H_0 as true
 - d. Reject H_0 ; the true mean lifetime differs from the claim
- 59 A hypothesis test about a nutritionist's claimed mean calorie count produces a p -value of 0.112. Using a significance level of $\alpha = 0.01$, state the correct conclusion.
- a. The test is inconclusive and must be repeated
 - b. Fail to reject H_0 ; not enough evidence that the true mean calorie count differs from the claim
 - c. Reject H_0 ; the true mean calorie count differs from the claim
 - d. Accept H_0 as true
- 60 A hypothesis test about a claimed mean test score produces a p -value of 0.089. Using a significance level of $\alpha = 0.05$, state the correct conclusion.
- a. Accept H_0 as true
 - b. Fail to reject H_0 ; not enough evidence that the true mean test score differs from the claim
 - c. Reject H_0 ; the true mean test score differs from the claim
 - d. The test is inconclusive and must be repeated

- 61** A hypothesis test about a claimed mean reaction time produces a p -value of 0.034. Using a significance level of $\alpha = 0.05$, state the correct conclusion.
- a. The test is inconclusive and must be repeated
 - b. Accept H_0 as true
 - c. Fail to reject H_0 ; not enough evidence that the true mean reaction time differs from the claim
 - d. Reject H_0 ; the true mean reaction time differs from the claim
- 62** A hypothesis test about a claimed mean reaction time produces a p -value of 0.089. Using a significance level of $\alpha = 0.01$, state the correct conclusion.
- a. Reject H_0 ; the true mean reaction time differs from the claim
 - b. Fail to reject H_0 ; not enough evidence that the true mean reaction time differs from the claim
 - c. Accept H_0 as true
 - d. The test is inconclusive and must be repeated